

Latent Trajectories in Recurrent Depth: Geometry, State-Tracking, and Formatting

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Abstract

We present a reproducible inference-only pipeline for analyzing reasoning trajectories in recurrent-depth transformers. We demonstrate that multi-step logic does not induce looping, refuting naive depth-encoding hypotheses. Instead, we find that steps-to-settle serves as an effective compute proxy strongly correlated with state retention. Furthermore, we reveal that two-scale orthogonality dynamics previously reported in the literature are methodology-dependent artifacts. Finally, we identify a massive formatting confound in capability evaluations and establish a timing hinge showing that answers are fixed well before geometric regimes become measurable.

1 Introduction

Each token’s latent path falls into one of several regimes—settling, looping, or drifting—distinguishable via Lyapunov exponents, recurrence, and persistent homology signatures. We evaluate three hypotheses regarding the relationship between trajectory shape and reasoning:

- H1 **Settle / loop / drift.** All answer-token trajectories exhibit exponential contraction. Unbounded drift is excluded by construction.
- H2 **Loops encode reasoning depth.** When the path loops, its winding number grows with the number of reasoning steps required.
- H3 **Contraction prevents state tracking.** If the update strongly contracts, it cannot maintain a running count.

1.1 Related Work

Recent work explores recurrent depths in language models. Geiping et al. introduced Huginn, observing looping on specific tokens. Blayney et al. found non-fixed-point behavior in a small fraction of tokens. Pappone et al. proposed two-scale orthogonal dynamics. Merrill et al. and Grazzi et al. highlighted limitations of state-tracking in linear and finite-precision recurrent models.

2 Methods

All experiments use the open Huginn-3.5B model in inference-only mode. Our pipeline extracts hidden states across the unroll dimension at the final layer.

Table 1: Experimental Datasets

Task	Levels	N (trajectories)	Prompt format
PARARULE-Plus	depths 2–5	10 / depth	"Question: ... Answer:"
Counting (± 1)	lengths 2–32	5–15 / len	"Start at 0. Add 1. Subtract 1..."
Parity (switch)	lengths 4–48	10 / len	"Light is off. Flip. Wait..."
Running-max	lengths 4–48	8 / len	"Numbers: 3 7 2 ... Largest?"
FineWeb	–	1000	–

2.1 Metrics Implemented

- **Rotation power:** Fraction of non-DC power in the period-6 bin over a 24-unroll tail (5280-D).
- **Winding number:** Accumulated angle in 2D-PCA projection, normalized to 2π .
- **Steps-to-settle:** First step where $\|\Delta h_t\| < 0.1$, serving as a proxy for effective computation.
- **Persistent homology H1:** Maximum normalized persistence (Rips complex).

3 Results

3.1 Geometric Regimes and The Formatting Confound

Using the rotation statistic, we find that a single instruction word switches the recurrent map between a settling regime and a damped period-6 rotation, causally controlled by the token’s embedding row.

Behavioural attempts to connect this structure to capability revealed a massive evaluation confound. Scoring the first generated token, the model answers correctly on 12.5% of items. Reading the generated text, however, the answer is present in 78.1%. The model is not failing to compute the answer; it is failing to put the answer where standard metrics look. Supplying the output format recovers the published benchmark number.

Measuring the rotation statistic over two distinct windows separates two facts: over the decision window (unrolls 0–11) rotating and settling prompts are indistinguishable, but over the tail they separate completely. Because the answer is best-ranked at a median unroll of four, behavioural nulls about the regime manipulate a property that has not yet come into existence when the readout is settled.

3.2 E1: PARARULE-Plus (H2 Refuted)

All 40 answer-token trajectories (depths 2–5) exhibit settling behavior. Correlation analysis reveals that winding vs depth (per-row, $N=40$) yields $\rho = -0.037$, $p = 0.82$. Multi-step symbolic logic does not induce looping; winding-depth correlation is null.

3.3 E2: Synthetic Counting (H3 Supported)

Running ± 1 sum (length $n_{\text{ops}} \in \{2 \dots 32\}$): Steps-to-settle vs n_{ops} yields $\rho = 0.718$, $p < 1e - 5$. The model fails tasks requiring state retention beyond a few operations, confirming theoretical limitations.

3.4 E3: Length-Matched Control

To isolate state-tracking from prompt length, E3 used *Track* and *Local* tasks of identical length. *Track* yields steps vs n_{ops} correlation of $+0.558$, while *Local* yields -0.576 . The sign of the correlation is determined solely by task type: state retention increases computation.

3.5 E4: Phase Map and Force-Loop

With reduced budget ($num_steps=16$), loops appear (7/8 loop at $n_{ops} = 24$). With $num_steps \geq 24$, all trajectories settle. Loops are a symptom of insufficient computation budget.

3.6 E5: Generalization to Parity

Consistent with counting, steps-to-settle correlates with parity sequence length ($\rho = 0.615$).

3.7 Two-Scale Dynamics on Synthetic Data

Applying Pappone et al.’s two-scale metrics to synthetic data, we find that Acceleration grows with depth ($4.89 \rightarrow 7.45$, $\rho = 0.997$), providing a strong signal where Winding fails.

Table 2: Two-scale dynamics on synthetic data (Depths 2-5)

Depth	Accel (mean $\pm$ std)	Orthogonality ($\cos \theta$)	Settled
2	4.89 ± 0.02	-0.497 ± 0.002	50%
3	5.84 ± 0.01	-0.498 ± 0.001	50%
4	6.70 ± 0.02	-0.498 ± 0.001	50%
5	7.45 ± 0.02	-0.498 ± 0.002	50%

3.8 Reproduction of Orthogonality on Real Data

During reproduction on FineWeb text data, we obtained results that fundamentally differ from Pappone et al. We found that without step normalization, orthogonality is confounded with step norm (correlation -0.0436 after normalization). Instead of spiral dynamics, we observe oscillatory dynamics ($\cos \approx -0.687$).

4 Discussion and Conclusion

Answer-token trajectories in Huginn-3.5B converge to fixed points. We detect no winding-depth signal, refuting naive H2. However, steps-to-settle acts as an effective compute proxy, strongly supporting H3. Furthermore, two-scale orthogonality is shown to be a methodological artifact, and evaluating reasoning capability is heavily confounded by the model’s formatting willingness.

Contribution

Component	Alexandr Shianov	David Sverdlov	Arsenii Varaksin
Extraction pipeline	✓	–	–
Winding & Steps-to-settle	✓	–	–
PARARULE & Counting exps	✓	–	–
Length-matched controls	✓	–	–
Phase map experiments	✓	–	–
Two-scale dynamics	–	✓	–
FineWeb orthogonality repro- duction	–	✓	–
Formatting confound & regime timing	–	–	✓

Alexandr Shianov (45% of work): Developed extraction infrastructure, MVP metrics, and conducted E1–E5.

David Sverdlov (35% of work): Implemented two-scale dynamics metrics and conducted FineWeb reproduction.

Arsenii Varaksin (20% of work):

- Discovered the formatting evaluation confound.
- Defined and validated the scalar `rotation_power` statistic.
- Established the timing hinge separating regime from answer computation.